

# SatNOGS Audio to Operational Telemetry

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## BEACON DECODER

GNU Radio | G3RUH AX.25 | CRC-valid 263-byte frames

**5 / 5**

observations passed

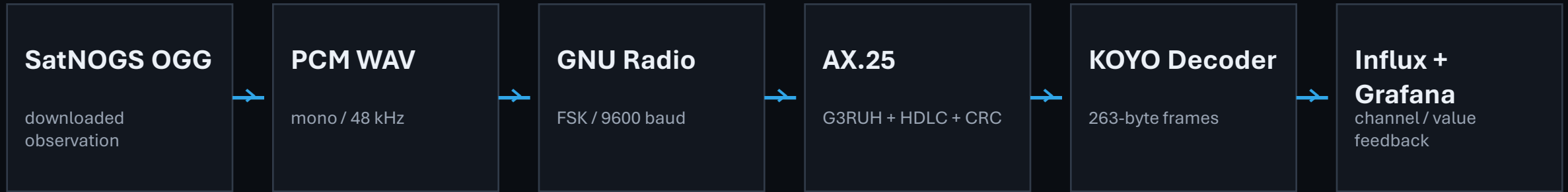
**22**

valid local frames

**20**

byte-exact control matches

# End-to-End Architecture



## Decoder input

The OGG recording is the source. SatNOGS demodulated frames are used only as an independent validation control.

**Store-and-forward: newest available upload, not continuous live RF.**

# Mission Data Coverage

55

UTC days with frames

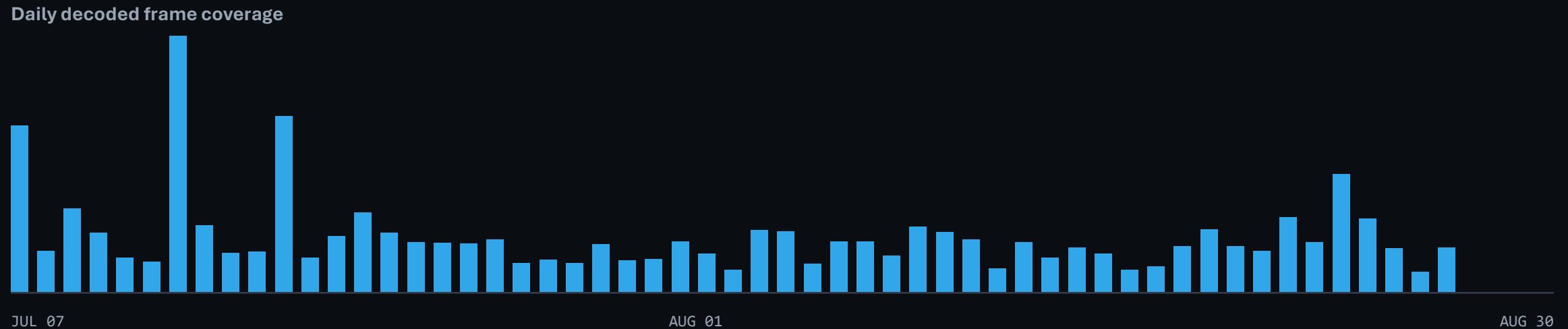
1,055

SatNOGS observations / 114 stations

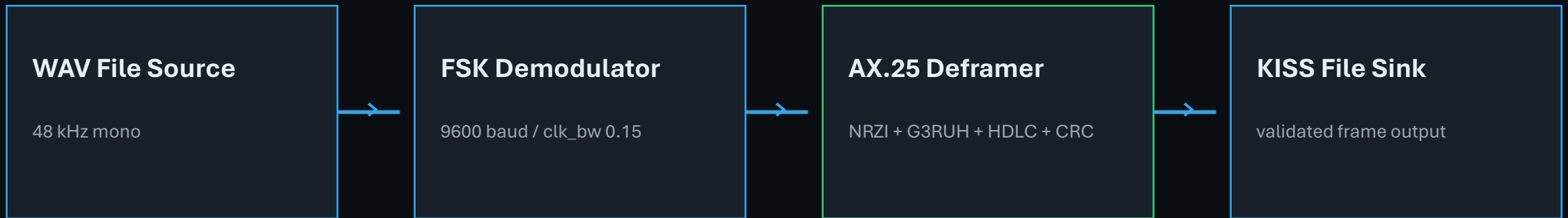
17,155

decoded frames

2026-07-07 to 2026-08-30 UTC



# GNU Radio Receive Flowgraph

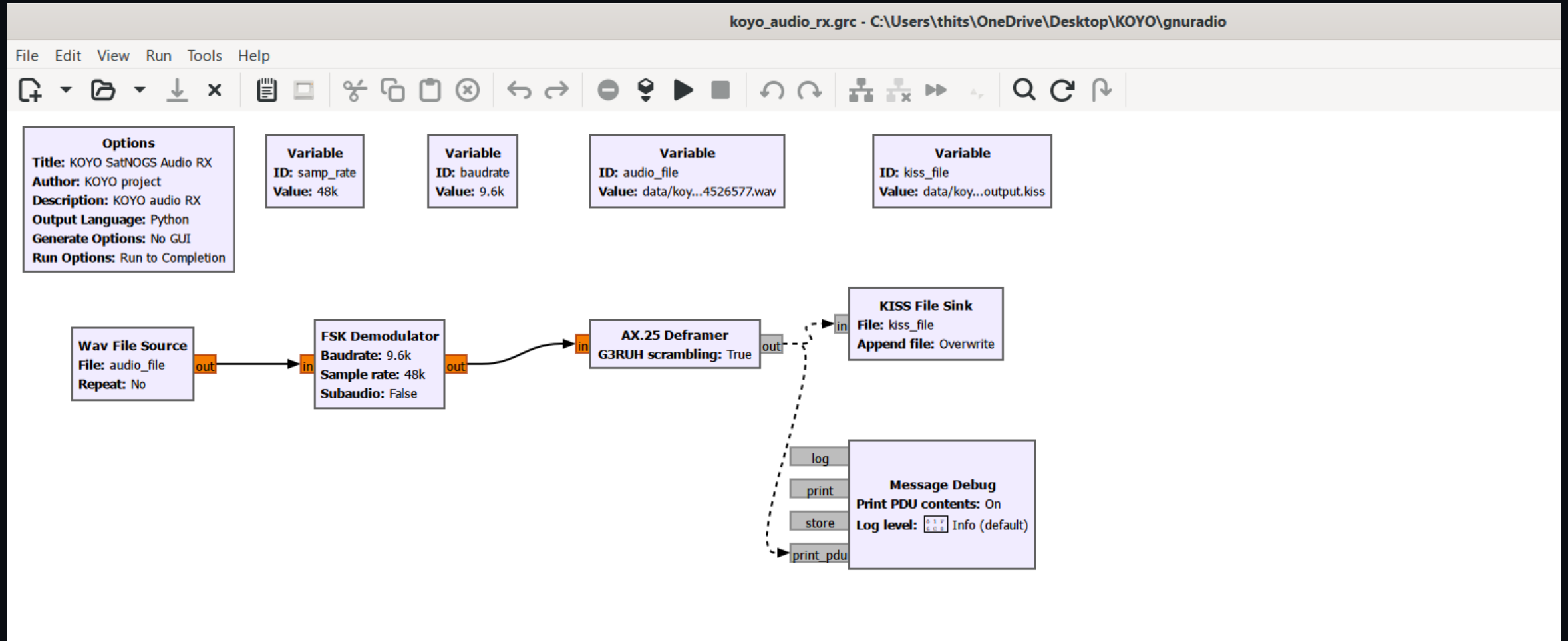


Editable flowgraph

`gnuradio/koyo_audio_rx.grc`

Observed malformed or short PDUs stay diagnostic. Only CRC-valid 263-byte KOYO frames reach telemetry decoding.

# GNU Radio Companion - Actual Flowgraph



# Audio Validation Across Stations

| OBSERVATION | STATION     | OFFICIAL | LOCAL | EXACT | RECOVERY | RESULT |
|-------------|-------------|----------|-------|-------|----------|--------|
| 14526577    | MAUSyagi    | 35       | 1     | 1     | 2.9%     | PASS   |
| 14637273    | EA3AGB      | 3        | 1     | 1     | 33.3%    | PASS   |
| 14909294    | MAUSyagi    | 31       | 14    | 14    | 45.2%    | PASS   |
| 14909617    | W6MSU UHF   | 4        | 4     | 3     | 75.0%    | PASS   |
| 14909703    | MAUSyagi-AK | 1        | 2     | 1     | 100.0%   | PASS   |

Overall exact recovery: 20 / 74 = 27%

Not decoded = not recovered from OGG by current settings, or rejected diagnostic PDU; it does not automatically mean an invalid spacecraft frame.

# Operational Dashboard

<🕒2026-07-07 00:00:00 to a few seconds agoUTC>🔍🔄Refresh30s v

LATEST MEASUREMENT (UTC)  

2026-08-30 18:42:21

SPACECRAFT MODE  

NOT DECODED

MODE FIELD NOT MAPPED

BOOT COUNTER  

22

BATTERY VOLTAGE  

NOT DECODED

VOLTAGE FIELD NOT MAPPED

BATTERY CURRENT  

NOT DECODED

CURRENT FIELD NOT MAPPED

ORBIT

LIVE ORBIT - KOYO

• KOYO

Altitude — km

Speed — km/s

Sub-lat —

Sub-lon —

UTC —

LIVE - position real (SGP4)

◀ LocateTrack: Off

• LIVE▶ SIM⌚ TIME

Real-time SGP4 from Koyo's TLE.

drag to orbit · scroll to zoom

BEACON DECODER STATUS

DECODER STATUS  

PASS

OBSERVATION  

14909703

STATION

2026-08-30 18:42:06MAUSyagi-AK

CAPTURED KISS  

6

VALID KOYO  

2

EXACT MATCHES  

1

RECOVERY  

100%

43 panels | decoder status + raw HEX | confirmed, candidate, and NOT DECODED states remain explicit

07

# Morning Demo and Next Step

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## ONE-COMMAND DEMO

```
powershell -ExecutionPolicy Bypass -File .\\local-stack\\live_refresh.ps1
```

### What it demonstrates

1. Fetch newest uploaded SatNOGS OGG
2. Decode locally with GNU Radio
3. Keep CRC-valid 263-byte frames
4. Push telemetry to InfluxDB and Grafana

### Current boundary

SatNOGS is store-and-forward.  
Unconfirmed engineering mappings stay candidate.  
Next: validate remaining fields against an authorized reference.

**Result: reproducible audio-to-dashboard chain with traceable validation evidence.**